

Pranav Vivek Malpure

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EDUCATION

- University of California San Diego** (Sept'24 - Mar'26)
Master of Science in Electrical and Computer Engineering | Intelligent Systems, Robotics & Controls **GPA: 3.64/4**
Courses: Statistical Learning-I, Introduction to Robotics, Linear Systems Theory, Sensing/Estimation in Robotics, Linear Algebra, Visual Learning, Planning/Learning in Robotics
- Indian Institute of Technology Bombay, India** (Nov'20 - Aug'24)
Bachelor of Technology with Honors, Aerospace Engineering **GPA: 8.06/10**
Minor in Systems & Controls Engineering
Courses: Navigation & Guidance of UAVs, Embedded Robotics, Reinforcement Learning, Intelligent Feedback & Control
Achievements: Ranked 1981 in India out of 250,000 candidates in the Joint Entrance Examination(Advanced) (2020)

WORK EXPERIENCE

Robotics Software Engineer | Labelbox | San Francisco, CA (Jul'25 - Present)

- Designed and implemented a modular **robot-learning framework** from scratch supporting interchangeable policies, robots, controllers, dataloaders, and logging systems for rapid experimentation across manipulation research workflows
- Trained and deployed $\pi 0$ with **LoRA** on in-house bimanual teleoperation datasets; achieving successful bowl grasping, picking and stacking behaviors in **live deployment** on physical xArm6 robots
- Fine-tuned and trained **R3M** vision encoder from **scratch** on proprietary egocentric datasets; achieved **25.3%** task success on Relocate task in Adroit sim vs. **16%** baseline with open-source data, validating dataset quality
- Built end-to-end **VR** based teleoperation infrastructure for bimanual arms using **ROS2** & real time robot control pipelines
- Developed end-to-end egocentric data post-processing pipeline; converting diverse capture formats (wrist, overhead stereo) into MCAP, adding action labels, and QAing pilot deliverables for customers

Robotics Intern | Flytbase Labs | Pune, India (Jun'23 - Jul'23)

- Optimized** real-time addition of NFZs resulting in **reduction** of computing time by **92%** by grouping visibility graphs
- Developed an algorithm that assesses reachability of subsequent waypoints online and optimizes return-to-home decisions

KEY PROJECTS

Sample Efficient RL for Dexterous Manipulation | ERL, UC San Diego (Oct'24 - Present)
Graduate Student Researcher

- Trained an expert SAC policy for a **22-DOF** xArm6-Allegro robot to pick up a cube via staged reward shaping and curated about **10k** high-quality demonstration trajectories
- Fine-tuned ACT-based policies with online RL methods combining behavioral cloning and demonstration replay for dexterous manipulation tasks
- Built a hardware abstraction layer for the physical Allegro robot, bridging the ManiSkill API to **real-world execution**
- Worked on implementing **3D diffusion** policy for combining 3D data and denoising actions trained on imitation learning

Perception based Pedestrian Intent Prediction | UC San Diego (Apr'25 - Present)

- Developed a pedestrian intent prediction model achieving up to **88%** F1 score utilizing VGG-16 for feature extraction and a Convolutional LSTM for spatio-temporal dynamics
- Boosted prediction accuracy by using a **learning rate scheduler** & experimenting with different input sequence lengths
- Enhanced temporal analysis by integrating pedestrian bounding box and YOLO-Pose derived body pose data into a novel LSTM-based architecture for binary intent classification

The Humanoid Project | Student Tech Team, IIT Bombay (Mar'22 - Apr'24)
Team Lead

- Led a team of **20** students building a full sized humanoid robot to be deployed for **sorting** books in the central library
- Crafted roadmaps to ensure technical coordination between subsystems & oversaw budget allocation of INR **0.2 million**

Autonomous Navigation of UUVs | Aerospace Dept., IIT Bombay (Jan'23 - Apr'23)

- Implemented the **curvature velocity method** in python to navigate a UUV through static obstacles using ROS-Gazebo
- Leveraged data from **3** onboard **sonar** sensors to detect obstacles, enabling **real-time** adjustments of thrust & velocity

TECHNICAL SKILLS

Languages/Frameworks C++, Python, MATLAB, Robot Operating System (ROS), ROS 2, Git, Embedded Linux

Packages and Libraries Numpy, Pandas, SciPy, pyvisgraph, Pytorch

Softwares and Simulators Gazebo, RViz, dm_control-MuJoCo, Maniskill-SAPIEN